Unified Military Analytics and Comparison Dashboard – DV

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1. Project Overview

The **Unified Military Analytics and Comparison Dashboard (DV)** is a fully integrated data analytics platform designed to analyze global military power in 2025.

The system collects, processes, and visualizes military and economic indicators for more than 140 countries using automated data pipelines and interactive dashboards.

The project combines:

- Web Scraping (Python)
- Data Cleaning & Transformation
- KPI Engineering
- Power BI Dashboard Development
- Interactive Navigation & Filtering
- Documentation & GitHub Packaging

The final output is an integrated analytics ecosystem that enables:

- Global military trend analysis
- Country-level intelligence profiling
- Side-by-side country comparison
- Coalition power simulation

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2. Tech Stack

Area	Tools Used
Scraping	Python, requests, BeautifulSoup
Processing	pandas, numpy
KPI Engineering	Python (custom calculations)
Visualization	Power BI Desktop
Documentation	Markdown, GitHub

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3. Milestone 1 – Data Collection & Preparation

Objective

To scrape structured military and economic metrics from GlobalFirepower.com for 140+ countries.

3.1 Web Scraping Implementation

A dictionary was created mapping URLs to metric names.

```
Dict = {
    'https://www.globalfirepower.com/total-population-by-country.php':
'total_population',
    'https://www.globalfirepower.com/available-military-manpower.php':
'total_military_manpower',
    ...
}
```

Each webpage was parsed using:

- `requests`
- `BeautifulSoup`
- Regular expressions for numeric cleaning

```
response = requests.get(u)
soup = BeautifulSoup(response.text, 'html.parser')
```

The scraped values were cleaned using regex:

```
regex = "[^0-9]"
value = (re.sub(regex, "", value))
```

3.2 Data Structuring

The data was stored in long format and converted into wide format using pivot:

```
df = df.pivot(index="country", columns="metric", values="value")
```

Output

- `military_raw_data.csv`
- Structured dataset for 140+ countries

- 50+ military and economic indicators

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4. Milestone 2 – KPI Engineering & Metadata Enrichment

Objective

To transform raw military metrics into meaningful analytical KPIs.

4.1 Power Score & Ranking

```
df['Power_Score'] = df['active_personnel'] + df['tanks'] +
df['total_military_aircraft'] + df['total_naval_fleet']
df["Power_Rank"] = df["Power_Score"].rank(ascending=False, method="min")
df["Power_Index_Rank_Gap"] = df["Power_Rank"] - 1
```

This metric evaluates relative military strength across nations.

4.2 Assets per Capita

```
df["Total_Assets"] = df["tanks"] + df["total_military_aircraft"] +
df["total_naval_fleet"]
df["Assets_Per_Capita"] = df["Total_Assets"] / df["total_population"]
```

This measures military asset intensity relative to population size.

4.3 Budget-to-GDP Ratio

```
df["Budget_to_GDP_Ratio"] = df["defense_budget_usd"] /
df["purchasing_power_parity_usd"]
```

This shows defense spending burden relative to economic strength.

4.4 Metadata Engineering

- NATO vs Non-NATO classification
- Continent mapping

- Region grouping

```
df["Alliance"] = df["country"].apply(
    lambda x: "NATO" if x in nato_countries else "Non-NATO"
)
```

Output

- `military_final.csv`
- Cleaned dataset ready for Power BI
- 5+ engineered KPIs

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5. Milestone 3 – Dashboard Development

Four fully integrated dashboards were developed in Power BI.

5.1 Global Overview

Provides a high-level executive summary including:

- Total Countries Analyzed
- Global Population
- Total Military Assets
- Power Rank Gap Distribution
- NATO vs Non-NATO Distribution
- Budget-to-GDP Global Map

Purpose: Strategic macro-level understanding.

5.2 Country Intelligence Profile

Allows deep dive into individual country performance.

Includes:

- Manpower Strength
- Air Power Distribution
- Naval Strength
- Land Power Map
- Defense Budget Analysis

Purpose: Detailed country-level analysis.

5.3 Power Comparison

Enables side-by-side comparison of two selected countries.

Features:

- Manpower Comparison
- Defense Budget Comparison
- Power Rank Comparison
- Aircraft Strength Analysis
- Population Comparison

Purpose: Bilateral strategic evaluation.

5.4 Coalition Strength Simulator

Simulates alliance-based military aggregation.

Features:

- Multi-country selection
- Coalition vs Reference comparison
- Budget Difference Analysis
- Naval Fleet Strength Difference
- Aircraft Strength Difference

Purpose: Strategic alliance modeling.

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6. Navigation Implementation (Power BI)

Navigation was implemented using:

Insert → Buttons → Navigator → Page Navigator

This allows seamless switching between:

- Global Overview
- Country Analysis
- Power Comparison

- Coalition Simulator

All filters and slicers were linked across dashboards.

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7. Key Achievements

- Successfully scraped 140+ countries
- Engineered 5+ custom KPIs
- Built 4 integrated dashboards
- Implemented dynamic filters and slicers
- Designed navigation system
- Developed coalition simulation logic

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8. Final Deliverables

- military_raw_data.csv
- military_final.csv
- Power BI Dashboard (.pbix)
- GitHub Repository
- Complete Documentation

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9. Conclusion

The Unified Military Analytics and Comparison Dashboard successfully integrates data engineering, KPI modeling, and interactive visualization into a single analytical ecosystem.

The platform enables users to:

- Evaluate global military power
- Compare nations strategically
- Simulate coalition strength
- Analyze defense-economic relationships

The project demonstrates strong skills in:

- Python Data Engineering
- KPI Modeling
- Power BI Dashboard Development
- Analytical Thinking
- Documentation & System Integration